

Fully branch-wise DTL rates on the archaeal big tree: likelihood landscape, best fit, cross-validation, and a reproducible protocol

gpurec / AleRax comparison — This_study / Undine C60

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Abstract

We re-analyse the rooting of the archaeal “big tree” (This_study / Undine C60: 257 taxa, $S = 513$ species-tree branches, 7,059 gene families given as ALE conditional-clade distributions) with **gpurec**, a GPU re-implementation of AleRax’s duplication–transfer–loss (DTL) gene-tree/species-tree reconciliation. **gpurec** reproduces AleRax’s per-family log-likelihoods to Pearson $r = 0.999998$. Allowing the DTL rates to vary *freely along every branch* ($\sim 2,000$ free parameters) recovers the same **Euryarchaeota** root as AleRax and the same ancestral (LACA) gene content, with duplication/transfer rates that are essentially clade-constant and loss rates that are branch-heterogeneous. The free per-branch model is the highest-likelihood fit found, and held-out cross-validation shows its advantage *generalises* (it is not overfitting). The likelihood surface has (at least) three basins; the global optimum (Euryarchaeota) is reached by warm-starting from a clade-grouped fit, but is missed by every cold (uniform) initialisation we have tried. We document the reproducible protocol and the open cold-start problem.

1 Setup and engine fidelity

The species tree has $S = 513$ branches; the data are 7,059 gene families (**.ale** conditional-clade distributions built from UFBoot samples). Everything is in $\log_2$ (bits) space internally; rates reported here are natural rates, and log-likelihoods are in *nats* (natural log) unless noted. Fraction-missing is handled in the AleRax-faithful **e-only** mode.

Engine fidelity. Evaluated at AleRax’s *exact* per-branch rates, **gpurec**’s per-family log-likelihoods match AleRax’s (**per_fam_likelihoods.txt**) to **Pearson** $r = 0.999998$ for every candidate root and every AleRax model class (**global**, **DTL_br2**, **DTL_br1.0**), up to a constant per-family offset. The engine is therefore not in question; the discussion below concerns the *optimiser* and the *model richness*.

2 The likelihood landscape: three basins

Fitting the rates by maximum likelihood reveals a multi-modal surface. Table 1 summarises the three basins we observe (total data log-likelihood, natural log, Eury-rooted unless noted). The deciding coordinate is the *origination* distribution O (where gene families are born on the tree): a deep, concentrated O gives Euryarchaeota; a diffuse O gives a near-tie or the small-genome-attraction (SGA) artifact.

Basin	data $\log L$ (nats)	how reached
Deep / Euryarchaeota (best)	-1,631,537	warm-start from clade-grouped fit
Cold near-tie	-1,664,248 (Eury)	cold global-rate warm-up
Cold SGA stall	$\approx -1,712,000$	cold uniform initialisation

Table 1: Three basins of the big-tree likelihood. The deep/Euryarchaeota basin is the global optimum found, ~ 33 k nats above the cold near-tie and ~ 80 k above the cold SGA stall. In the cold near-tie basin the SGA root Cluster2 (-1,663,618) marginally *edges* Eury (-1,664,248); only in the deep basin does Eury win decisively. The initialisation selects the basin.

Why cold fails. At full ML the deep basin is the global maximum, but it has a *small basin of attraction*: from a uniform initialisation the optimiser converges to the near-tie or SGA basin and never climbs to the deep one. The deep basin requires the origination O and the DTL rates to be jointly near the AleRax solution — they are coupled — which is why a joint warm-start works and partial seeds do not (§6).

3 The best fit: fully branch-wise (FULLbasin)

The best fit is a *fully free per-branch* model: every branch has its own duplication, loss and transfer rate plus its own origination weight ($\sim 2,000$ free parameters), warm-started from the clade-grouped fit. It attains $\log L = -1,631,537$ nats, the highest found, ~ 11 k nats above the clade-grouped fit.

Rooting. Euryarchaeota is rank 1 by +3,568 nats over the next root; the deep clades {Eury, TackA, DPANN} sit at the top and the small-genome roots at the bottom (Cluster2 is -12,190 nats below Eury).

AU test. A Shimodaira approximately-unbiased (AU) test over all 15 candidate roots, on the free per-branch per-family likelihoods, places **Euryarchaeota alone** in the 0.05 confidence set (BP=1.0); every other root, *including MHH* (-1,637,704, rank 4, -6,167 nats behind Eury), is rejected (AU ≈ 0). By contrast AleRax’s parsimonious DTL_br2 model gives a *region* {Euryarchaeota, MHH} (MHH only 9.7 nats behind). The free model is therefore *sharper*; we read this as the consequence of model richness plus the warm-start, not as stronger evidence (see §4 for why it is nonetheless not overfitting).

Rate structure (Fig. 1). With rates free to vary, duplication and transfer come out essentially *clade-constant* — clade membership (the paper’s 23 DTL classes) explains $R^2 \approx 0.94$ of their variance — whereas loss is *branch-heterogeneous*: only $R^2 \approx 0.38$ of its variance is clade-level. Most of the free model’s ~ 11 k-nat advantage over clade-grouping lives in the loss rates.

Origination. The fitted O is concentrated but free: two branches carry $\sim 32\%$ of the origination mass (top branch 0.19), over a near-uniform diffuse floor (73 branches hold 50%, 279 hold 90% of the mass).

Ancestral genome (LACA, Fig. 2). Of the 7,059 families, the free model places **903** at the root and the clade-grouped (AleRax) rates **772** (thresholded at posterior ≥ 0.5); **722** are shared

(Jaccard 0.76), and the per-family origination-at-root probabilities correlate $r = 0.96$. The free model reconstructs a somewhat larger ancestral genome.

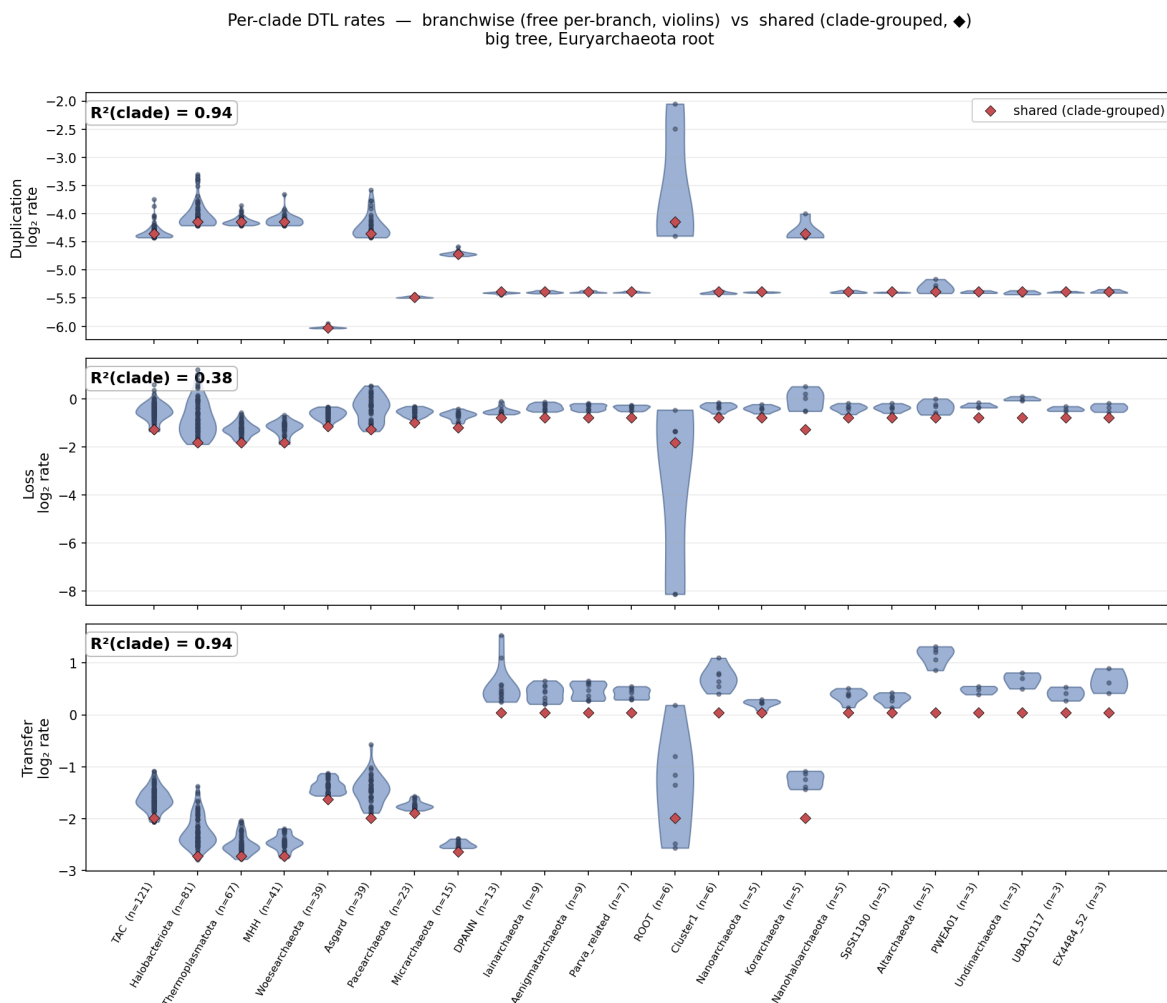


Figure 1: Per-clade DTL rates: branchwise (free per-branch, blue violins) vs. shared (clade-grouped, red ◆), one panel per rate. D and T violins are tight around the shared rate ($R^2 \approx 0.94$); L violins are wide ($R^2 \approx 0.38$).

4 Cross-validation: the richness is not overfitting

To test whether the $\sim 2,000$ free parameters fit noise, we fit both the clade-grouped and the free per-branch model on a *training half* (first 3,530 families) and scored the *held-out half* (3,529 families never used in fitting).

Scope. This establishes that the per-branch *rate* richness generalises (the crux of the overfitting concern). It does not separately cross-validate the AU *region* (which would require per-root refits). The index split is also family-size unbalanced (train $-389/\text{fam}$ vs. test $-73/\text{fam}$), so this is a generalisation test *across* a family-size shift; a random split would give a cleaner balanced number.

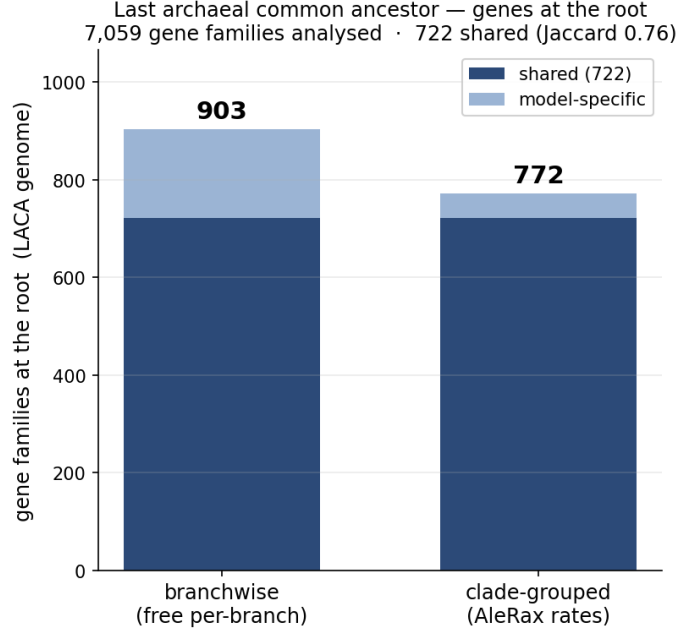


Figure 2: Gene families at the root (LACA): 903 under the free per-branch model vs. 772 under clade-grouped (AleRax) rates; 722 shared (Jaccard 0.76), of 7,059 analysed.

model	train $\sum \log L$	held-out $\sum \log L$	train/fam	held-out/fam
free (per-branch)	-1,373,618	-257,725	-389.13	-73.03
grouped (clade)	-1,383,170	-259,280	-391.83	-73.47
free — grouped	+9,551	+1,556		+0.44

Table 2: Held-out cross-validation (Eury root). The free per-branch model beats the clade-grouped model on *held-out* families by +1,556 nats (+0.44/family). Overfit parameters would hurt out-of-sample; these help. The train advantage (+9,551) shrinks to +1,556 held-out — $\sim 5/6$ of the in-sample gain is optimism, but the remaining positive sixth generalises. **Verdict: not overfitting.**

5 Reproducible protocol (the minimal robust setup)

All commands run from the `gpurec-cpp` checkout on a single A100 (`largegpu`). Paths use the cluster layout; `$U` is the output directory, `$DD` the data root, `$CG` the AleRax clade-grouping directory.

```

U=/work/SzollosiU/gergely-szollosi/williams_run/undine
DD=/work/SzollosiU/gergely-szollosi/williams_run/data/3_Reconciliation/This_study
CG="$U/alerax_ref/3_Reconciliation/This_study/5_reconciliation_models/DTL_br1_0/
↳ Undine_C60_Euryroot2_OR"
# data: tree $DD/4_species_tree/Undine_C60_<ROOT>root_short_name.nw
# .ale $DD/3_UFBOTs/ufboot_for_alerax/*.ale (7059 families)
# fraction_missing $DD/fraction_missing

```

Step 1 — clade-grouped warm-up (lands in the deep basin). Fits the paper’s clade-grouped DTL plus origination, initialised from AleRax’s rates. This is the step that selects the deep/Euryarchaeota basin. (~ 30 min on one A100.)

```
python experiments/run_undine_branchwise.py --root Eury \
--families 0 --min-species 1 --fm-mode e-only --prior none \
--clade-groups "$CG" --init-from-alerax --origination optimize \
--family-batch-size 500 --steps 250 --dtype float64 \
--out $U/AXgrp_DTL_br1_0_Eury.rates.txt
```

Step 2 — fully free per-branch fit (the best fit). Warm-started from Step 1; no rate grouping, no prior, free per-branch origination. (~1–7.5 h on one A100, depending on convergence.)

```
python experiments/run_undine_branchwise.py --root Eury \
--families 0 --min-species 1 --fm-mode e-only --prior none \
--origination optimize \
--init-from-rates $U/AXgrp_DTL_br1_0_Eury.rates.txt.json \
--family-batch-size 500 --steps 250 --dtype float64 \
--out $U/FULLbasin_Eury.rates.txt
```

Step 3 — rooting / AU over all 15 roots. Run Steps 1–2 for each root (substitute `--root` and the matching Undine_C60_<ROOT>root2_OR grouping dir), then:

```
python experiments/au_fullbasin.py # AU test on the FULLbasin per-root likelihoods
```

Step 4 — ancestral genome (LACA) content.

```
python experiments/laca_compare.py --root Eury \
--sidecar $U/FULLbasin_Eury.rates.txt.json \
--out $U/laca_Eury.json
```

Step 5 — overfitting cross-validation. Fit on the training half (`--families 3530`) for both models, then score held-out:

```
# grouped on train half
python experiments/run_undine_branchwise.py --root Eury --families 3530 \
--min-species 1 --fm-mode e-only --prior none --clade-groups "$CG" \
--init-from-alerax --origination optimize --family-batch-size 500 \
--steps 250 --dtype float64 --out $U/CVgrouped_Eury_train.rates.txt
# free per-branch on train half, warm-started from the grouped train fit
python experiments/run_undine_branchwise.py --root Eury --families 3530 \
--min-species 1 --fm-mode e-only --prior none --origination optimize \
--init-from-rates $U/CVgrouped_Eury_train.rates.txt.json \
--family-batch-size 500 --steps 250 --dtype float64 \
--out $U/CVfree_Eury_train.rates.txt
# held-out verdict
python experiments/cv_eval.py --root Eury --train-n 3530 \
--free $U/CVfree_Eury_train.rates.txt.json \
--grouped $U/CVgrouped_Eury_train.rates.txt.json \
--out $U/cv_eval_Eury.json
```

Key flags. `--fm-mode e-only` (AleRax-faithful fraction-missing); `--prior none` (no Brownian/TV smoothing; add `--dtl-tv-lambda <λ>` for a fused-lasso that auto-groups the free rates); `--origination optimize` (fit per-branch origination; the rooting signal); `--clade-groups <dir>` + `--init-from-alerax` (clade-grouped model seeded from AleRax — the basin selector); `--init-from-rates`

<sidecar> (warm-start chain); --family-batch-size 500 (memory; the $S=513$, 7059-family forward must be batched); --dtype float64 (gradient accuracy); --families N (use the first N families; 0 = all).

6 Open problem: cold-start basin-finding

The deep/Euryarchaeota basin is the global optimum but is only reached by warm-starting (Steps 1–2). From a cold (uniform-origination) start, every strategy tried so far lands in the near-tie or SGA basin:

- anti-concentration (Simpson) origination barrier — directionally *wrong* (pushes O toward uniform; sinks Euryarchaeota);
- fixed root-mass / depth origination priors — over-constrain and destroy the fit;
- global-rate warm-up — reaches only the near-tie (−1,664k);
- fused-lasso (TV) annealing — regularises the DTL, not the (bistable) O ; stuck at the near-tie;
- structured- O seed ($\omega \propto -\text{depth}$) — diffuses straight back to the near-tie.

The deep basin needs the structured O *and* the matching DTL together; a cold, initialisation-independent route to it is unsolved. The most promising untried idea is a *deep-origination prior homotopy*: impose a strong root-region origination prior to put the optimiser in the deep basin, then anneal the prior to zero (the unconstrained optimum it then converges to should be the global/deep basin). This is the directionally-correct counterpart of the failed anti-concentration barrier.